MEDICATION SAFETY SIGNAL MONITOR · HAND-IN R00

# Start here

Version 1.0, 5 November 2026: what this is, what to read first, and what the words mean

| What this is | What this is not |
|---|---|
| • The front door to the hand-in set<br>• One glossary, one code key, one evidence legend | • A results report (the evidence is in R01 to R08)<br>• Clinical advice: a signal is a lead, not proof of harm |

**How to read this.** Read sections 1 and 2 first. Keep sections 3 to 5 open as look-up tables while you read the other reports. Section 6 and the dark box are the facts to remember.

## 1 · What this project is

| | In one line |
|---|---|
| The user and her questions | **Dana**, a hospital pharmacy safety officer (fictional persona, R04 ch. 2), asks each morning: (1) Which drugs have side effects reported that the label does not mention? (2) Which have an active recall? (3) What has changed since yesterday? |
| The trust rule | Every flag traces to its source: the label versions searched (DailyMed) and the FDA recall record open in one click; the FAERS case reports are reachable by query, not yet from her screen. |
| The 13 drugs | **8 opioids:** oxycodone, hydrocodone, morphine, fentanyl, hydromorphone, methadone, buprenorphine, tramadol. **5 GLP-1 drugs** (diabetes and weight loss): semaglutide, liraglutide, dulaglutide, tirzepatide, exenatide. Exenatide has no reference label, so signals exist for 12 of 13 (D018, D031 v5). |
| Data and platform | Public data only, no patient records. Built on **Palantir Foundry** with its AI layer **AIP**. The app is one Workshop module of 4 pages (v0.9.0). |

## 2 · The hand-in set, in the order to read it

| # | Read | It answers |
|---|---|---|
| 1 | Architecture note: picture and build facts | What is the whole system? (Skip its decision boxes at first.) |
| 2 | R07 Data judgment, chapter 1 | What would messy public data do to the officer's morning? |
| 3 | R04 The application, chapters 1–4 | How does the officer use the app, and what does each number mean? |
| 4 | R01 Foundry from scratch | What is Foundry, which parts did we use, and where do I click? |
| 5 | R02 Pipelines: raw to ontology | How does raw public data become signals, recalls and changes? |
| 6 | R03 The Ontology (ch. 6 = NAION trace) | How does a flag link back to its source? |
| 7 | R06 Optimization · R08 AI reliability · R07 rest | What did we measure? How often is the AI wrong? Which data problems did we fix or leave? |
| 8 | R05 Decisions, and what we did not do | Why each choice, and what we left out. Read last, as the index. |
| + | Deck and R09 Demo navigation guide | What the 20-minute demo shows, click by click, and what each stop proves. |

Files: `docs/11-handin/architecture-note/architecture-note.pdf` · `docs/11-handin/deck/MSSM-demo-deck.pdf` · `docs/11-handin/reports/pdf/R01…R09`. Order from `docs/11-handin/reviews/first-time-reader.md` §6.

## 3 · Glossary

| Term | Plain meaning (Foundry and general) | Term | Plain meaning (drug safety) |
|---|---|---|---|
| **Palantir Foundry** | A data platform holding data, the code that cleans it, a model of real things and the apps, with version history. | **FAERS** | FDA Adverse Event Reporting System: voluntary reports of suspected side effects, quarterly. Moved to **AEMS** in March 2026 (DQ20). |
| **Dataset** | A versioned table in Foundry. Ours run raw → clean → curated. | **Case** | One FAERS report of one patient event; only its latest version counts (D022). |
| **Transform** | Python code that reads datasets and writes one. One run = a *build*. | **Suspect drug** | A drug the reporter suspects (primary, secondary, interacting); drugs only *also taken* do not count (D021). |
| **Code Repository** | A git repository in Foundry holding the transforms. | **MedDRA PT** | Preferred term: the standard word for a reaction. Its groupings need a licence we lack. |
| **Data Lineage** | The graph of which dataset feeds which, with build status. | **2×2 table** | For a drug and reaction: *a* = both; *b* = drug, other reactions; *c* = other drugs, this reaction; *d* = the rest. |
| **Ontology** | Foundry's layer of objects, links and Actions on curated datasets. | **PRR** | Proportional reporting ratio = (a ÷ the drug's reports) ÷ (c ÷ other drugs' reports): how much more often the reaction comes with this drug. |
| **Object (type)** | One real thing / its kind (Drug); one dataset row each. | **Evans rule** | A common screen: 3+ reports, PRR ≥ 2, chi-squared ≥ 4 (D006, proposed). |
| **Link type** | A named relationship between two object types, e.g. Signal → Drug. | **DailyMed / SPL** | The official US label site / its XML format (Structured Product Labeling). Our label truth (D019). |
| **Action** | A form that writes to the Ontology: ours saves a Triage Decision (*write-back*). | **Label section** | A titled part of a label. We search Boxed Warning, Contraindications, Warnings (and Precautions), Adverse Reactions. |
| **Workshop** | Foundry's point-and-click app builder; Dana's screen is one. | **openFDA** | The FDA's public web API: recall, Drugs@FDA and NDC data. |
| **AIP** | Palantir's AI Platform: language models in pipelines. Ours reads label sections. | **Recall** | FDA enforcement report of a product withdrawn or corrected; Class I most serious. *Active* = Ongoing. |
| **F1** | One test score balancing *found every right item* (recall) and *added no wrong ones* (precision). 1.0 is perfect. | **RxNorm / RxCUI** | US standard drug vocabulary / its concept id. Our drug key = ingredient RxCUI, resolved once (D020). |
| **NAION** | Sudden sight loss from optic-nerve damage; semaglutide × NAION is our worked example. It is a *known positive*, not a discovery: EMA PRAC (6 Jun 2025) and a WHO alert (27 Jun 2025) named it; our 3 US labels do not mention it (DECISIONS learning 2026-11-05). | **NDC** | National Drug Code: the US code for one package. |
| | | **Reference label** | The label versions searched for a drug (Drug → Products → label). |

## 4 · Key to the ID codes

| Code | Means | Look it up in |
|---|---|---|
| D### | A decision (where the brief is silent). `D016 v3` = third version; old ones kept. 12 are still *proposed* (D006, D015, D025–D029, D037, D045–D048), many already built; the owner accepts them in one batch. | `docs/06-decisions/DECISIONS.md` (D001–D048) |
| A### | An approach we tried, with its verdict: kept, dropped or parked. | `docs/05-approaches/INDEX.md` (A000–A017) |
| DQ## | A data-quality issue found in the public data. | `docs/07-data-quality/data-issues-register.md` (DQ01–DQ42) |
| O# | An optimization, measured before and after (O1, O2, O3, O3a). | `docs/08-optimization-log/optimization-log.md` |
| G1–G6 | Gate defaults in the dataset contracts, accepted via D039. | `docs/04-architecture/dataset-contracts-v1.md` |
| Step N | 12-step build order; each step ends with a proof (*gate*). 6–8 = first slice; 9 = all 13 drugs; 10 = incremental FAERS; 11 = screen + health checks; 12 = hand-in. | `docs/04-architecture/foundry-conventions/project-build-order-v1.md` |

| Code | Means | Look it up in |
|---|---|---|
| Baseline gate | The full rebuild where every hand-in number was frozen. | `docs/11-handin/baseline-gate.md` |
| R## | A hand-in report, R00 (this page) to R09. | `docs/11-handin/reports/pdf/` |
| 30–38 | Diagrams: 30 data flow (v2), 31 ontology, 32 signal pipeline, 33 incremental (v2), 34 change feed, 35 monitoring, 36 demo map, 37 ripple map, 38 one-page architecture. | `docs/diagrams/README-handin.md` |

## 5 · Evidence labels and status bands

Each claim carries an evidence label (what kind of claim); each chapter-closing box a status band (how far along). Text read live from `tools/handin/style.py`, so it matches every report.

| Evidence | Meaning | Status | Meaning |
|---|---|---|---|
| OBSERVED | Read directly from a public source (FAERS, DailyMed, openFDA, RxNorm), with the query that produced it. | TESTED | Built, run, and checked against evidence. |
| DERIVED | Computed by our pipeline from observed data, for example a PRR or a count per drug and reaction. | PLANNED | On the plan; not built yet. |
| ASSUMED | Not verified: taken as true for now because the data or the brief is silent. It could be wrong. | PROPOSED | Suggested; waiting for a decision. |
| DESIGNED | A design choice we made and wrote down (the decision log says why). It may be built; the status band says how far. | | |

## 6 · Five numbers and what they mean

| Number | What it means | Do not read it as | |
|---|---|---|---|
| **1,952** signals | Drug–reaction pairs that meet the Evans rule **and** whose reaction is not found in any reference label of the drug (1,844 *not in label*) or only via a medium-band synonym (108 *possibly in label*). | Proven side effects. It means more reports than expected, not cause. | DERIVED |
| **1,073** ranked | The **review queue**: signals *not in label*, in the clinical bucket (D016 v3), under half from mass reports of 30+ drugs (D023 v2). 1,355 at step 9; synonyms and buckets cut it. | 1,073 findings. | DERIVED |
| **126** active recalls | openFDA recalls with status **Ongoing**, out of 900 recalls pulled for our ingredients. | *Not terminated* (that would add Completed). | OBSERVED |
| **26** changes | Change events (signal, recall, label) in the **last 30 days** (Morning brief card). Page 3's date range gives *since my last review* (D045, proposed). | Changes since yesterday. | DERIVED |
| **F1 0.97** | AI label reading on a 35-item answer key (Boxed Warning + Contraindications of 2 oxycodone labels; written by Claude, frozen before the run, no pharmacist yet); found 35 of 35 items; target 0.85 (D015, proposed). Separately **448/448** extracted terms are grounded (quote found) over all 9 sections. | A score for all 448 terms, or for any GLP-1 label. | DERIVED |

**What this means for Dana (the safety officer)**

**FAERS is about 130 days behind.** Signals change only when a new FAERS quarter arrives (here 2026Q1-2026Q2), so *what changed* is mostly recalls and labels.

**FAERS moved to AEMS in March 2026** (DQ20). Where future quarters will come from is not verified yet.

**Only the project owner can submit a Triage Decision today** (R03 ch. 5); giving Dana that right is a production step.

**The AI's quote is not on her screen** (R04 ch. 7); a *not in label* signal has nothing to quote anyway (R03 ch. 6).

**The hero drug had 0 AI-read labels.** The AI read only 2 oxycodone labels, so NAION is *not in label* from the exact-text and synonym checks on 3 semaglutide labels.

**The Morning brief *changes* card is the last 30 days**, not since yesterday. Use page 3's date range.

**Gate: How we know this page is right** TESTED

**Required evidence:** All 24 numbers are `docs/11-handin/numbers.json` lookups (captured 2026-11-05T07:50Z; keys listed in this page's source file); definitions copied from the Phase D fix brief; each code names a real file.

**Result:** Met: builds with no layout problems; every page looked at.

**If it fails:** Fix this page first: the other reports rely on its meanings.